

EVALUATION



CHAPTER Nine

Instructor Name :- Suzana Binti Ahmad

Software testing

- Software testing is the process of evaluating and verifying that a software product or application does what it is supposed to do.

Types of Software testing

- There are many different types of software test
 - Acceptance testing – verify whether the whole system works as intended
 - Integration testing – ensuring that software components of functions operate together
 - Unit testing – validating that each software unit performs as expected. A unit is the smallest testable component of an application
 - Functional testing – checking functions by emulating business scenarios, based on functional requirements. Black-box testing is a common to verify functions

Types of Software testing

- There are many different types of software test
 - Performance testing – testing how the software performs under different workloads.
 - Regression testing – checking whether new features break or degrade functionality
 - Stress testing – testing how much strain the system can take before it fails
 - Usability testing – validating how well a customer can use a system to complete a task

Usability testing

- Usability testing is the practice of testing how easy a design is to use with a group of representative users.
- It usually involves observing users as they attempt to complete tasks
- Often conducted repeatedly, from early development until a product's release

Usability testing

- Process
 - Plan – define, decide what you want to test
 - Set user tasks – prioritize the most important tasks; clearly define tasks with realistic goals; create scenarios where users can try to use the design naturally
 - Recruit testers – know your target group
 - Facilitate/moderate testing – set up testing in a suitable environment. Observe and interview

Heuristic evaluation

- A heuristic evaluation is a usability inspection method for computer software that helps to identify usability problems in the user interface (UI) design.



Heuristic evaluation

- It specifically involves evaluators examining the interface and judging its compliance with recognized usability principles.



Heuristic evaluation

- The main goal of heuristic evaluations is to identify any problems associated with the design of user interfaces.



Heuristic evaluation

- Usability consultant Jakob Nielsen developed this method on the basis of several years of experience in teaching and consulting about usability engineering.



The 10 heuristic evaluation

- The 10 most general principles for interaction design. They are called "heuristics" because they are more in the nature of rules of thumb than specific usability guidelines.



The 10 heuristic evaluation

1. Visibility of system status (Feedback)

- The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.

Upload the file or [Cancel](#)



1.0 Base Camp by 37 signals

The upload button is enabled, until clicked.

Then it is replaced with a progress indicator until the file has finished uploading

The 10 heuristic evaluation

1. Visibility of system status (Feedback)

- The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.

Type new password:

Six-characters minimum; case sensitive

Password strength: Strong

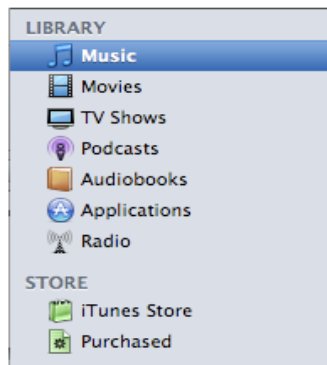
1.1 Windows Live Account

Password strength is shown as the password is entered

The 10 heuristic evaluation

2.Match between system and the real world (Metaphor)

- The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.



2.0 iTunes

Organized as a library that contains your media library: music, movies, shows, audiobooks. Beneath the Library is the Store where you can buy more media to put in your Library.

The 10 heuristic evaluation

3. User control and freedom

- Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.



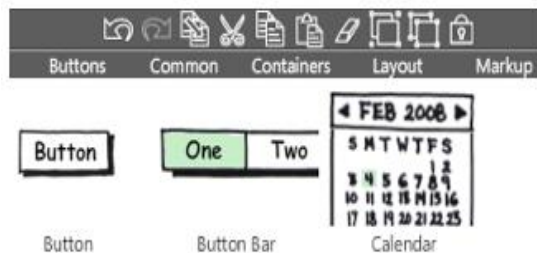
3.0 CollabFinder

Search is easy to open, enter info, execute or cancel.

The 10 heuristic evaluation

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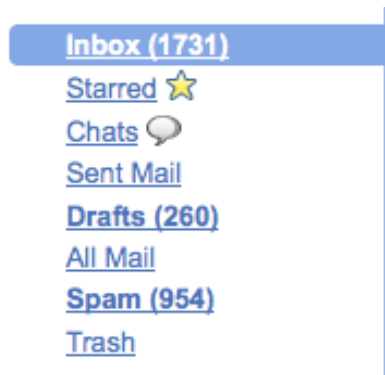
3.1 Balsamiq

Undo and Redo buttons are available in the toolbar, and can also be accessed with the standard keyboard shortcuts

The 10 heuristic evaluation

4. Consistency and standards

- Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.



4.0 Gmail

When Gmail was designed, they based the organizational folders on the same ones used in client email applications: Inbox, Drafts, Sent Mail.

The 10 heuristic evaluation

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- Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.



4.1 Microsoft Office

Word, Excel, and PowerPoint all use the same style toolbar with the same primary menu options: Home, Insert, Page Layout... Consistency results in efficiency and perceived intuitiveness.

The 10 heuristic evaluation

5. Error prevention

- Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.



5.0 Yammer

Disables the update button after it is clicked, so the person cannot update the post twice by accident

The 10 heuristic evaluation

5. Error prevention

- Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.



A screenshot of a Google search interface. The search bar contains the text 'design'. Below the search bar, a list of auto-suggested search terms is displayed, each followed by the number of results. To the right of the suggestions, there are links for 'Advanced Search', 'Preferences', and 'Language Tools'. At the bottom right of the suggestions list is a 'close' link.

design	
design within reach	5,350,000 results
designer handbags	3,430,000 results
designer shoes	2,630,000 results
designer clothes	3,120,000 results
designer dresses	1,110,000 results
design sponge	9,930,000 results
designer	265,000,000 results
design museum	13,600,000 results
designers guild	630,000 results
designer jeans	2,010,000 results

5.1 Google Auto Recommend

The auto recommend feature cuts down on mis-spellings

The 10 heuristic evaluation

6. Recognition rather than recall

- Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.



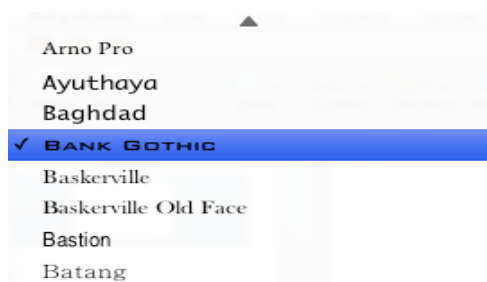
6.0 Quanta IDE

Type ahead for coding in a development environment

The 10 heuristic evaluation

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6.1 Keynote

Previews the fonts you can pick from, instead of just the font name

The 10 heuristic evaluation

7. Flexibility and efficiency of use

- Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

Common Shortcuts

Add Action	Return
New Window	⌘N
Synchronize with Server	⇧⌘S
Clean Up	⌘K
Planning Mode	⌘1
Context Mode	⌘2
Inbox	⇧⌘1
Quick Entry	⇧⇧Space

Quick Entry's shortcut can be customized in Preferences

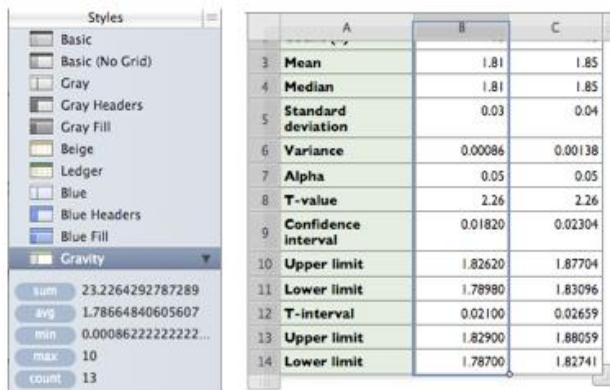
7.0 OmniFocus

List of keyboard shortcuts and accelerators

The 10 heuristic evaluation

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The screenshot shows the 'Styles' sidebar on the left with various formatting options like 'Basic', 'Gray', 'Blue', etc. The main area displays a spreadsheet with columns A, B, and C. The data is as follows:

	A	B	C
3	Mean	1.81	1.85
4	Median	1.81	1.85
5	Standard deviation	0.03	0.04
6	Variance	0.00086	0.00138
7	Alpha	0.05	0.05
8	T-value	2.26	2.26
9	Confidence interval	0.01820	0.02304
10	Upper limit	1.82620	1.87704
11	Lower limit	1.78980	1.83096
12	T-interval	0.02100	0.02659
13	Upper limit	1.82900	1.88059
14	Lower limit	1.78700	1.82741

Below the spreadsheet, a summary statistics panel shows:

- sum: 23.2264292787289
- avg: 1.78664840605607
- min: 0.0008622222222...
- max: 10
- count: 13

7.1 Numbers- Apple's Spreadsheet product
Previews common function results on the left when a column is selected, more efficient that clicking on an action in the toolbar

The 10 heuristic evaluation

8. Aesthetic and minimalist design

- Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility. Visual layout should respect the principles of contrast, repetition, alignment, and proximity.

The 10 heuristic evaluation

8. Aesthetic and minimalist design

8.0 Kontain

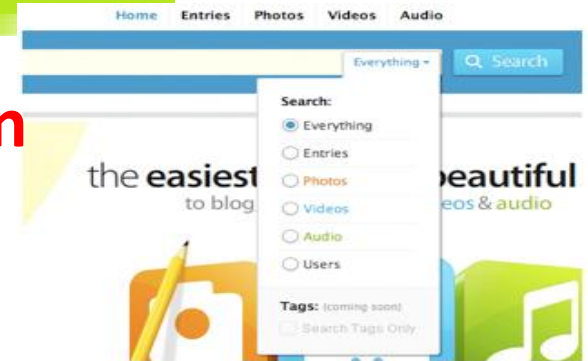
Kontain' search menu exemplifies the four principles of visual design:

Contrast: bold text is used for the two labels in the search

Repetition: the orange, blue, and green text match the media types

Alignment : strong left alignment of text, right aligned drop down

Proximity: a light rule is used to separate tags from the other options



The 10 heuristic evaluation

9. Help users recognize, diagnose, and recover from errors

- Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

Or start a new account

Choose a username (no spaces)

bert

⚠ bert is already taken. Please choose a different username.

Choose a password

⚠ Passwords must be at least 6 characters and can only contain letters and numbers.

Retype password

Email address (must be real!)

not an email

⚠ The email provided does not appear to be valid

☒ Send me occasional Digg updates.

9.0 Digg

Provides immediate feedback with specific instructions

The 10 heuristic evaluation

9. Help users recognize, diagnose, and recover from errors

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9.1 Humorous 'Page Not Found' Error

Uses a funny image and copy, but provides viable alternatives (article listings and blog link) and a course of action (report it)

The 10 heuristic evaluation

10. Help and documentation

- Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.



10.0 Picnik

Contextual help (this is an example of help in the 'Collages' module) tips in Picnik are clear and easy to navigate

The 10 heuristic evaluation

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10.1 GoodBarry

Embedded videos can be used to showcase features as well as get people started using the product

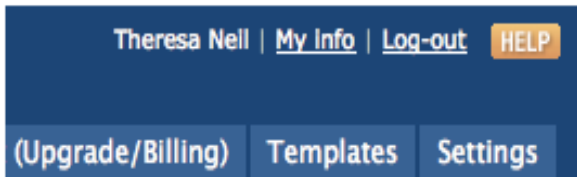
The 10 heuristic evaluation

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10.2 BaseCamp by 37signals

Help opens a new browser window/tab with a full set of help resources: search, FAQ, video tutorials, customer forums





Happy



Neutral



Worried

HOW to conduct a heuristic evaluation

- In general, heuristic evaluation is difficult for a single individual to do because one person will never be able to find all the usability problems in an interface.





Happy



Neutral



Worried

HOW to conduct a heuristic evaluation

- Luckily, experience from many different projects has shown that different people find different usability problems.





Happy



Neutral



Worried

HOW to conduct a heuristic evaluation

- Therefore, it is possible to improve the effectiveness of the method significantly by involving multiple evaluators.



How?



- It is certainly true that some usability problems are so easy to find that they are found by almost everybody, but there are also some problems that are found by very few evaluators.



How?



- First, it is not necessarily true that the same person will be the best evaluator every time.



How?



- Second, some of the hardest-to-find usability problems are found by evaluators who do not otherwise find many usability problems.



How?



- Therefore, it is necessary to involve multiple evaluators in any heuristic.



How?



- The recommendation is normally to use three to five evaluators since one does not gain that much additional information by using larger numbers.



How?



- Heuristic evaluation is performed by having each individual evaluator inspect the interface alone. Only after all evaluations have been completed are the evaluators allowed to communicate and have their findings aggregated.



How?



- This procedure is important in order to ensure independent and unbiased evaluations from each evaluator.



How?



- The results of the evaluation can be recorded either as written reports from each evaluator or by having the evaluators verbalize their comments to an observer as they go through the interface.



How?



- Written reports have the advantage of presenting a formal record of the evaluation, but require an additional effort by the evaluators and the need to be read and aggregated by an evaluation manager.



How?



- Using an observer adds to the overhead of each evaluation session, but reduces the workload on the evaluators.



How?



- Also, the results of the evaluation are available fairly soon after the last evaluation session since the observer only needs to understand and organize one set of personal notes, not a set of reports written by others.



Example



System's Title: Adapting Meta-Cognitive as an Educational Tool: Animated Puzzle

Cosmetic: Need not be fixed

Minor: Needs fixing but low priority

Major: Needs fixing and high priority

Catastrophic: Imperative to fix

Interface and Interactivity

No	Heuristic Evaluation	Cosmetic	Minor	Major	Catastrophic	Possible solutions
1	The interface is user friendly	☐	☐	☐	☐	
2	Match between system and the real world (metaphor)	☐	☐	☐	☐	
3	User control and freedom	☐	☐	☐	☐	
4	The color used is excellent and interesting	☐	☐	☐	☐	

Example



5	Consistency and standards (color, button, font)	☐	☐	☐	☐	
6	Animation used is suitable for learning	☐	☐	☐	☐	
7	Recognition rather than recall	☐	☐	☐	☐	
8	Flexibility and efficiency of use	☐	☐	☐	☐	
9	Aesthetic and minimalist design	☐	☐	☐	☐	
10	The interactivity is very good and fun	☐	☐	☐	☐	

End of Lecture